Simple Wall Collider

Procedural Collider Generation with Splines

Overview

The **Simple Wall Collider** tool is a powerful Unity plugin designed for generating procedural wall-like colliders based on splines. It enables developers to create accurate and optimized colliders for fences, walls, barriers, and more. The tool supports mesh generation, ground projection, and visualization options for a seamless integration into your workflow.

Features

- **Spline-Based Collider Generation**: Generate colliders from splines dynamically.
- **Ground Projection**: Align spline vertices to terrain or other layers.
- **Mesh Simplification**: Optimize colliders with configurable simplification thresholds.
- **Invertible Normals**: Option to invert mesh normals for specific use cases.
- **Quality Control**: Adjustable quality settings for mesh precision.
- **Custom Height and Offset**: Control the height and base offset of the collider.
- **Gizmos Visualization**: Preview colliders in the editor with customizable gizmo colors.
- **Easy-to-Use Inspector**: Dedicated editor inspector for managing all tool settings.

Installation

1. Import the package into your Unity project.
2. Attach the `WallCollider` component to any GameObject.
3. Configure settings in the Inspector to suit your needs.

How to Use

1. Setup

- Add the `WallCollider` script to a GameObject

2. Generating Colliders

- Open the `WallCollider` Inspector.
- Create a spline
- Adjust the **Collider Preferences** (quality, height, offset, etc.).

- Note: The collider is automatically generated when the spline or tool preferences are changed. If it does not update, press the **Generate Collider** button.

3. Ground Projection

- To align the spline to the terrain or specific surfaces:
  - Configure the `Ground Projection Layer` in the Inspector.
  - Press the **Project On Ground** button.

4. Visualization

- Use the **Gizmos Draw Type** to toggle visualizations:
  - `None`: Disable gizmos.
  - `Selected`: Show gizmos when the GameObject is selected.
  - `Always`: Always display gizmos.
- Customize the gizmo color with the `Gizmos Color` property.

5. Setup for AI navigation

- If you are using NavMesh Agents or another AI navigation tool that operates with meshes, you can easily set up your wall colliders for compatibility. Simply press the **Configure for Nav Mesh** button in the inspector. This will add the necessary **MeshFilter** and **MeshRenderer** components to your GameObject, ensuring the generated mesh is ready for NavMesh baking or integration with other navigation systems.

Inspector Properties

Collider Preferences

- **Invert**: Invert the normals of the generated mesh.
- **Quality**: Controls the density of the mesh along the spline.
- **Base Offset**: Adjusts the vertical position of the collider.
- **Height**: Sets the height of the collider walls.
- **Simplification**: Reduces mesh complexity based on angle thresholds.
- **Ground Projection Layer**: Defines the layers to project spline vertices onto.

Visual Preferences

- **Gizmos Draw Type**: Select when gizmos should be visible.
- **Gizmos Color**: Sets the color for gizmos.

Known Limitations

- The tool relies on the Unity Spline package, which must be installed.
- Complex splines with high detail may impact performance.

Support

For bug reports, feature requests, or general support, contact us at:
vasylkhreptak12@gmail.com

License

This package is licensed for use within Unity projects. Redistribution or resale of the source code or binaries is prohibited.

Let me know if you'd like to modify or expand this documentation further!